

Jialun CAO

Research Assistant Professor

The Hong Kong University of Science and Technology

<https://jialuncaocao.github.io/>



Dear Members of the Selection Committee,

I am applying for the position of Tenure-track Assistant Professor at The School of Artificial Intelligence (SAI) of The Chinese University of Hong Kong, Shenzhen, drawn by the strong alignment between my academic background, research experience, and the role's responsibilities. Its distinguished reputation and its commitment to fostering an inclusive and innovative academic environment make it an ideal place to advance my career.

I am currently a Research Assistant Professor in the Department of Computer Science and Engineering at the Hong Kong University of Science and Technology (HKUST), where I also earned my Ph.D. Previously, I conducted collaborative research as a visiting scholar at the University of Stuttgart and CUHK (Shenzhen). I am also a mentee in the Asia Pacific Women in Leadership Program, mentored by a faculty member from Harvard University.

Research Experience – My research sits at the intersection of Software Engineering (SE), Artificial Intelligence (AI), and Formal Methods (FM), with a particular focus on harnessing advanced AI techniques to enhance software development, testing, and verification. The overarching goal of my research is to improve the reliability and development-efficiency of software systems, particularly intelligent and safety-critical systems. My research has been published in top-tier venues, including ICSE, ESEC/FSE, ASE, TOSEM, CAV, USENIX Security, AAAI, ACL, among others. To date, I have authored or co-authored 31 publications in these venues, with an additional 17 works currently under review. Of these, I have served as the first author on 11 papers (eight published, three under review), co-first author on three, and corresponding author on eight.

Teaching Experience – In Spring 2025, I taught COMP1021 Introduction to Computer Science at HKUST (110 students) as the instructor, receiving excellent teaching evaluation scores above both course and university-wide averages. I also served as a guest instructor for Nankai University's 2025 International Frontier Courses (220+ students). Previously, I was a teaching assistant for COMP1021 (Fall 2023) and COMP3021 Java Programming (Spring and Fall 2020), supporting lab tutorials, grading, and student consultations.

Area Recognition – My doctoral dissertation, "Towards Automatic Testing and Fault Localization in Natural Language Processing Systems", was recognized with the **ACM SIGSOFT Outstanding Doctoral Dissertation Award** for 2025, a prestigious award presented to only one Ph.D. dissertation in SE area worldwide per year. I am **the first Asian women recipient**. Additionally, my work received the ACM Distinguished Paper Award at ASE 2024.

Funding support – I have secured HK\$2.89 million (~\$370,000) in research funding as a Ph.D. student and postdoc and currently serve as Co-PI on a 2025/26 General Research Fund project (HK\$1,590,000) funded by Research Grants Council of Hong Kong, titled "A Multi-faceted Framework for Automated Pseudocode-to-Code Translation using Large Language Models." I have also established strong industry ties with ByteDance, Ant Group, and Huawei, and worked as a consultant to support industrial software development and quality assurance.

Academia Engagement – I actively contribute to the research community, serving on program committees for top-tier conferences such as ICSE, FSE, ISSTA and ASE, and as Session Chair at ESEC/FSE 2025 and ASE 2024. I have been invited to present my work at leading institutions (e.g., University College London, Peking University), CCF seminars, and industry forums. Most recently, I served as Local Chair for the 2025 Frontiers Forum on AI and Formal Methods, where distinguished speakers were invited, including a Turing Award laureate.

Looking ahead, I am dedicated to pursuing impactful research that bridges academia and industry. I believe my expertise and experience align well with the department's goals and research priorities, and I am enthusiastic about the opportunity to contribute to SAI's vibrant academic environment.

Enclosed for your consideration are my (1) CV, (2) Research Statement, (3) Teaching Statement, (4) Publication List, and (5) Teaching Evaluation, (6) Award/Degree Certificates.

I would be grateful if you would consider my application favorably!

Sincerely,
Jialun Cao